

BUILDING AN OPEN DATA INFRASTRUCTURE FOR INDIAN LANGUAGES

A Data Management Unit (DMU) Report • May 2022



Executive Summary

The National Language Translation Mission (NLTM), **Bhashini**, has been set up under the aegis of the Ministry of Electronics and Information Technology (MeitY) with the aim of making rapid progress in speech, text, and vision technology for Indian languages. The broader goal is to bring parity in AI technologies for Indian languages with respect to English through open-source contributions in tools, datasets, neural models, and reference applications.

This mission is highly cross-cutting in terms of supporting:

- All 22 constitutionally recognized languages in India.
- A wider variety of tasks including machine translation, speech recognition, speech synthesis, document character recognition, and language understanding.

AI4Bharat at IIT Madras serves as the Data Management Unit (DMU) for Bhashini and will help create open-source datasets across languages and tasks, often improving on existing datasets by orders of magnitude. Furthermore, AI4Bharat is developing an open-source tool for collaboration on language data collection and annotation.

Introduction

The National Language Translation Mission (NLTM), Bhashini, operates as a mission-mode project aimed at democratizing digital access by dismantling language barriers. AI4Bharat, a leading research center at IIT Madras with a singular mission of bringing AI language technology parity with respect to English, has been selected to serve as the Data Management Unit (DMU) of Bhashini.

As part of this role, AI4Bharat is constructing high-quality, open-source datasets for all 22 scheduled languages across multiple core AI tasks: Machine Translation (MT), Automatic Speech Recognition (ASR), Text-to-Speech (TTS), and Optical Character Recognition (OCR). The DMU provides the base layer of data infrastructure, allowing researchers, startups, and industry to build upon an open ecosystem. All datasets will be compliant with the Universal Language Collection API (ULCA) repository and released under permissible open licenses.

Challenges and Opportunities

Indian languages perfectly encapsulate the diversity of India alongside their rich, complex morphological structures. India is home to the fourth highest number of languages globally. Hindi, Urdu, Bengali, and Punjabi proudly sit in the top 20 most spoken languages across the world.

According to the 2011 census, there are 1,369 rationalized mother tongues (with over 10,000 speakers each), grouped into 121 languages. Crucially, 22 of these are constitutionally recognized (scheduled) languages. However, 191 mother tongues remain vulnerable or endangered. Furthermore, languages such as Sanskrit, Kannada, Telugu, Malayalam, Tamil, and Odia enjoy classical language status, reflecting their rich heritage and independent evolution.

This immense diversity, spanning multiple scripts, dialects, and accents, presents a profound challenge to the goals of **Digital India**. Ensuring that all citizens can access digital content and services seamlessly in their native language requires vast AI resources. A direct comparison between high-resource languages (like English or Spanish) and Indian languages reveals a stark readiness gap.

The Foundational Language Infrastructure

To achieve parity, there is an urgent need for a foundational language infrastructure layer acting as a public good. This addresses critical gaps by providing: (a) standardized tools for dataset creation, (b) large-scale diverse datasets for training AI, (c) state-of-the-art reference neural models, and (d) reference applications demonstrating these capabilities across diverse deployment scenarios (mobile, offline, low-power).

DMU's Mission

Given the linguistic landscape, the DMU's mission is to rapidly bridge the technological divide. The focus is to adopt best practices from high-resource language AI while ensuring that the richness, heterogeneity, and distinct morphological characteristics of Indian languages are faithfully preserved in digital ecosystems. The core goals include:

- **Cross-Country Collaboration:** Building a nationwide team of language experts spanning academia, the social sector, and professional agencies to collect diverse data at scale.
- **Unified Tooling:** Developing a robust, open-source tool for data collection that supports all 22 languages and standardizes workflows for ULCA compatibility.
- **Reference Implementations:** Publishing baseline models and applications for translation, transcription, and vision to spur commercial and academic innovation.

Current State of Data

This section outlines the baseline of available data across tasks and the strategic targets set by the DMU to elevate Indian language AI to state-of-the-art levels.

Machine Translation (MT)

Machine Translation requires massive parallel corpora (sentences paired perfectly across two languages). While recent efforts have aggregated millions of sentence pairs for dominant languages (e.g., Hindi, Tamil), there remains a critical deficit for low-resource languages such as Bodo, Dogri, Kashmiri, and Santali. The DMU's strategy involves heavy web mining, document alignment, and targeted human translation to establish robust, multi-domain parallel datasets for all 22 languages.

Automatic Speech Recognition (ASR)

ASR (Speech-to-Text) depends on thousands of hours of accurately transcribed audio. Current datasets are largely restricted to read-speech in quiet environments. To build robust models capable of understanding real-world, spontaneous speech, the DMU targets the collection of extensive audio data encompassing diverse genders, age groups, acoustic environments, and regional dialects.

Language Family / Region	Scheduled Languages Targeted	Primary Tasks
Indo-Aryan (Northern/Central)	Hindi, Urdu, Punjabi, Gujarati, Marathi, Sindhi, Nepali	MT, ASR, TTS, OCR
Indo-Aryan (Eastern)	Bengali, Odia, Assamese, Maithili	MT, ASR, TTS, OCR
Dravidian (Southern)	Tamil, Telugu, Kannada, Malayalam	MT, ASR, TTS, OCR
Sino-Tibetan & Austroasiatic	Bodo, Manipuri, Santali	MT, ASR (Foundational)

Text-to-Speech (TTS)

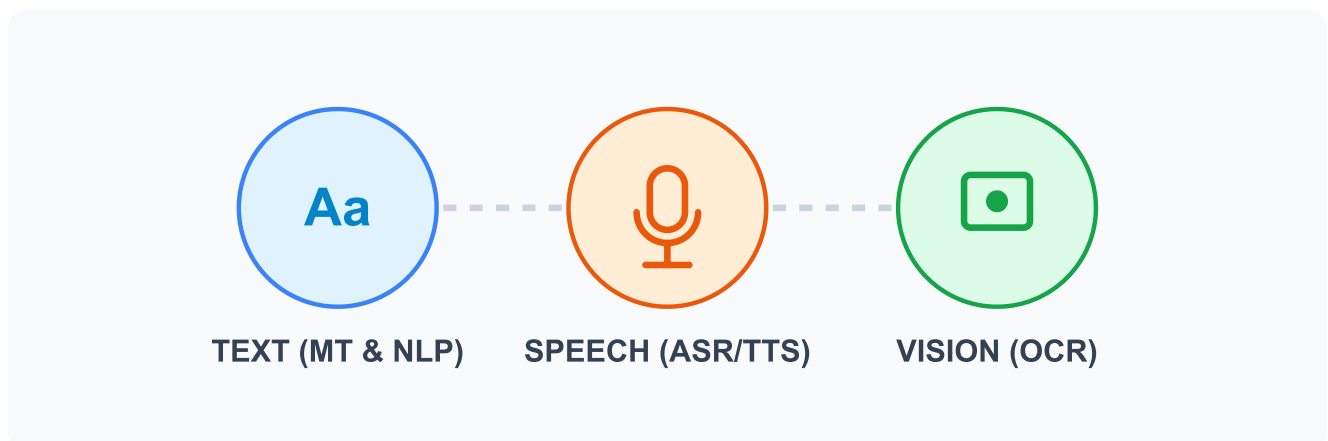
Unlike ASR, which requires diverse, noisy data from thousands of speakers, Text-to-Speech (Speech Synthesis) requires extremely high-fidelity, studio-quality recordings. The current state of open TTS data for Indian languages is highly fragmented. The DMU is coordinating professional studio recordings with dedicated native voice artists (one male and one female per language) to produce phonetically balanced datasets. This will enable the creation of natural-sounding, expressive AI voices for digital assistants and accessibility tools.

Document Optical Character Recognition (OCR)

Document OCR involves digitizing printed and handwritten text from scanned pages. While English OCR is highly accurate, Indian language scripts pose unique challenges due to complex conjuncts, modifiers (matras), and varied typographic standards. The DMU is compiling large datasets of scanned books, legal documents, and government gazettes, carefully annotated with word-level and line-level bounding boxes and their corresponding textual transcriptions.

Scene Optical Character Recognition

Scene text (text "in the wild" such as street signs, hoardings, product labels, and television tickers) requires separate AI models due to unpredictable lighting, perspective distortion, and varied fonts. The existing dataset landscape for Scene OCR in Indian languages is virtually non-existent. The DMU is pioneering efforts to capture and accurately annotate natural scene images across multiple states.



Domain Diversity Imperative

Language data must reflect real-world utility. The DMU strictly mandates data collection spanning multiple critical domains: daily news, legal proceedings, government announcements, agriculture, healthcare, women's empowerment, and digital payments. This ensures the resulting models serve grassroots digital inclusion.

DMU's Approach

To solve the challenge of resource limitation at a national scale, the DMU employs a multi-pronged approach that combines human expertise with scalable technology platforms.

1. Set up a cross-country team of language experts

Quality data requires deep linguistic insight. The DMU is establishing partnerships with academic institutions, state universities, and non-profit linguistic organizations across India. These teams provide the native-speaker oversight necessary for translation validation, audio verification, and complex orthographic quality control. By decentralizing the workforce, the project injects local dialectical nuances and cultural context that centralized data agencies often miss.

Dedicated project managers oversee these regional teams, ensuring adherence to the stringent guidelines defined by the DMU while maintaining steady throughput across all 22 scheduled languages.

2. Build a unified tool for data collection

Historically, language data collection relied on fragmented, disconnected tools, leading to inconsistent formats and lost metadata. To counter this, AI4Bharat is developing a singular, unified, open-source platform (internally referenced as *Shoonya*). This platform serves as the nerve center for all annotation tasks.

Multi-Task Workflows

Supports side-by-side translation, audio recording, audio transcription, and image bounding-box annotation within a single cohesive interface.

ULCA Standardization

Automatically formats all resulting datasets into the JSON-based MeitY Universal Language Collection API (ULCA) specification for immediate publication.

Multi-Tier Quality Assurance

Implements a maker-checker workflow where senior linguists review and grade the outputs of initial annotators before data is finalized.

Contributor Analytics

Real-time dashboards tracking hours recorded, sentences translated, and overall accuracy metrics to manage the nationwide workforce.

3. Mining Data at Scale

Manual human annotation is highly accurate but slow. To reach the required scale (millions of sentences), the DMU deploys automated web-mining pipelines. These tools continuously crawl public domains, identifying parallel document structures (e.g., a government report published in both English and Hindi). Advanced document parsing and sentence-alignment algorithms extract the raw data, which is then fed into the unified tool for human verification, significantly reducing the time required compared to translating from scratch.

4. Policy on sources to collect data from

Ethical data collection is paramount. The DMU operates under a strict policy framework:

- **Licensing:** Data is solely mined from sources that are in the public domain or explicitly licensed under open frameworks (e.g., Creative Commons, Govt Open Data License).
- **Privacy (PII):** Automated sanitization scripts scrub Personally Identifiable Information (names, phone numbers, addresses) prior to human review or dataset release.
- **Copyright:** No proprietary or paywalled content is ingested into the open infrastructure.

DMU's Timelines

The creation of this data infrastructure is executed in overlapping phases to ensure early wins while steadily building capacity for the most resource-starved languages.

Phase	Focus Area	Key Milestones & Deliverables
Phase 1 (Months 1-3)	Platform Setup & High Resource Initiation	Launch of unified annotation tool; Onboarding of initial expert teams; Mining parallel corpora for top 5 languages (Hindi, Tamil, Telugu, Marathi, Bengali).
Phase 2 (Months 4-6)	Speech Collection & Medium Resource	Commence large-scale ASR crowdsourcing; Expand MT efforts to 10 additional scheduled languages; Initial release of foundational baseline datasets.
Phase 3 (Months 7-9)	Full 22 Language Scale & TTS Production	Initiate data streams for low-resource languages (Bodo, Dogri, Santali); Complete professional studio recordings for TTS voices.
Phase 4 (Months 10-12)	Vision Tasks & Final Publication	Finalize Document and Scene OCR bounding box annotations; Comprehensive QA audit; Bulk upload of all datasets to the public ULCA repository.

Appendices: Data Collection Guidelines

To ensure consistency across thousands of annotators, the DMU enforces strict instructional guidelines for each task type. A summary of these guidelines is provided below:

A1. Guidelines for translating source sentences

Translators must capture the precise meaning of the source text without omitting or adding information. Cultural nuances and idioms must be localized appropriately rather than translated literally. Grammar, spelling, and punctuation must reflect standard written norms of the target language.

A2. Guidelines for collecting voice data from native speakers

For ASR, speakers must read naturally at conversational speed. Recordings must capture diverse acoustic environments (quiet rooms, slight background noise) and encompass a balanced demographic split (gender, age, geography). Audio clipping and extreme distortion must be rejected.

A3. Guidelines for transcribing audio data

Transcribers must provide an exact verbatim representation of the audio, including natural hesitations (e.g., "umm", "ah"), partial words, and restarts. Punctuation should be inferred based on speaker pauses and intonation. Numbers must be written as spoken (e.g., "twenty-two" vs "22").

A4. Guidelines for collecting voice data from professional artists (TTS)

TTS data requires absolute acoustic consistency. Artists must maintain a uniform distance from the microphone, consistent volume, and a neutral, professional tone across all sessions. Recordings must take place in an acoustically treated, soundproof studio environment using high-fidelity equipment.

A5. Guidelines for collecting image data for Optical Character Recognition

Images must be of high resolution, in focus, and legible to the human eye. Bounding boxes must tightly enclose the text at the line and word level without cutting off modifiers or descenders. For Scene OCR, annotations must note perspective skews and occlusions.

End of Report

For more information, visit <https://ai4bharat.iitm.ac.in/> or <https://bhashini.gov.in/>